

ELEC5305 Project Feedback Two

Wiener Speech Enhancement with VAD Guided Noise Updating

Yulong Chen | SID 550889472 | GitHub yche0441

Report updated 18 September 2026 | Experiment completed 13 September 2026

Project website: <https://ychen0441.github.io/elec5305-project-550889472/>

GitHub repository: <https://github.com/ychen0441/elec5305-project-550889472>

1. Objective and project progress

The research question is: **When does VAD-guided noise updating improve Wiener speech enhancement compared with a fixed initial noise estimate, and how does this depend on the time variation of the background noise?**

The first controlled experiment completed in MATLAB R2026a Update 5. It processed 48 clean/noisy input pairs and produced 144 rows of native SNR and STOI measurements. The result-collection correction 2026-09-13-r1 was used. Both the algorithm verification gates and the result-table regression check passed. The complete original run is preserved in the MATLAB result archive.

An existing implementation by LIU Ming / Pascal Scalart was reproduced before the controlled experiment. Its TSNR branch was adapted for the project. The two modes use a common analysis window, FFT, hop, SNR estimator, Wiener gain, initial PSD and reconstruction. Fixed mode keeps the initial noise PSD. VAD mode updates it recursively following non-speech decisions. The clean reference does not enter either enhancer.

The earlier proposal used independent environmental speech recordings. Following the teacher's feedback, NOIZEUS paired clean/noisy recordings now support the main quantitative test. The recordings are the corpus's prepared mixtures of speech and recorded real-world noise. They are not described as the student's own simultaneous recordings. A small selection of car and street conditions keeps the first experiment manageable. The short literature review connects eight papers to the design, including the existing TSNR implementation, VAD, noise tracking and intelligibility evaluation.

The first experiment shows small mean reference-SNR advantages from VAD updating but lower mean STOI than fixed estimation. This report presents the literature, reproducible method, complete condition comparisons, diagnostic evidence and remaining work. It is a progress report on the completed experiment; parameter optimisation, listener assessment and the final demonstration remain to be completed.

2. Literature review

Eight papers inform the experiment. The review covers established approaches relevant to this controlled comparison; it is not a comprehensive survey of modern speech enhancement. The distinction between published methods and the implementation used here is maintained throughout.

2.1 Wiener enhancement and the existing implementation

Scalart and Vieira Filho [1] examine frequency-domain enhancement and emphasize estimating the a priori SNR. This supports using an established SNR estimator when studying noise tracking, so the comparison does not simultaneously change several parts of the filter. The accessible abstract supports this overview; detailed claims about every equation or parameter in the 1996 paper are not made here.

Plapous, Marro and Scalart [2] discuss the delayed response of decision-directed SNR estimation and propose two-step noise reduction (TSNR), followed by harmonic regeneration noise reduction (HRNR). The supplied MATLAB function implements both branches. The present comparison uses only its TSNR branch, with the same gain calculation in fixed and VAD modes. HRNR would introduce an additional processing change and is left outside this first comparison. The paper provides the direct technical context for the reproduced source.

2.2 Voice activity and changing noise

Sohn, Kim and Sung [3] use statistical modelling, a likelihood-ratio decision and a hangover scheme to detect speech activity. This is a useful reference for why frame decisions and temporal continuity both matter. The project instead uses a simple energy ratio and five-frame hangover, as permitted by the teacher. It does not claim to reproduce Sohn's likelihood model. The simpler detector makes the interaction between missed speech and noise updating easier to inspect.

Martin [4] estimates noise from smoothed spectral minima without requiring a speech/non-speech detector. It supplies an alternative to hard VAD gating: background noise can be tracked using spectral history. Its treatment of smoothing and minimum-statistics bias also shows that noise estimation is more than selecting quiet frames. Implementing this alternative is outside the first controlled comparison, but it provides context if the energy detector fails in changing noise.

Cohen and Berdugo [5] use minima-controlled recursive averaging (MCRA). Their noise estimate depends on speech-presence information derived from a comparison with local minima. This supports recursive updating while showing why an uncertain decision may need more care than an immediate binary update. The project's update coefficient is fixed and its VAD is energy-based; this is not an implementation of MCRA.

Cohen [6] develops improved MCRA for adverse noise conditions. This extends the discussion from whether to update to how reliably speech absence can be inferred. Together, [4]-[6] motivate recording VAD errors and update activity alongside enhancement scores. They do not establish in advance that this project's simple VAD will outperform a fixed estimate.

2.3 Evaluation

Taal et al. [7] present STOI as a predictor of the intelligibility of noisy and time-frequency-processed speech. It complements reference SNR because waveform error and intelligibility are different properties. The experiment uses MATLAB's established stoi implementation, not a new approximation. A higher STOI score remains a prediction and does not replace listening. The paper's abstract and publication details, and the implementation documentation, were checked; a complete derivation review remains future reading.

Hu and Loizou [8] introduce a common noisy-speech corpus and compare enhancement algorithms using judgments that distinguish speech distortion, background distortion and overall quality. This supports using the same paired recordings for both methods and retaining listening examples. This small project does not reproduce their listener study or treat objective scores as listener ratings.

2.4 Implications for the experiment

These studies motivate keeping the TSNR path constant while changing only noise updating. Martin and Cohen provide alternatives to hard VAD gating, but implementing them would add further experimental variables. The energy detector is retained to examine whether its decisions help noise tracking or allow speech to enter the noise estimate. The results therefore test this specific implementation, rather than every adaptive noise estimator. Reference SNR, STOI, spectrograms and listening examples address different aspects of the outcome [1]–[8].

3. Data and experimental design

The supplied NOIZEUS archives contain 30 clean sentences and 240 corresponding noisy files for car and street noise at nominal 0, 5, 10 and 15 dB. All files passed the ZIP integrity check. WAVs are mono, 8 kHz, signed 16-bit PCM. Every corresponding pair has the same length, and a lag check over -8 to +8 samples found its strongest cross-correlation at zero lag. This limited check does not prove every possible aspect of dataset preparation.

For a small first experiment, the package includes six evenly spaced sentence IDs. Their speaker identifiers were subsequently confirmed in Tables 1-2 of the NOIZEUS paper.

Table 1. Sentence and speaker allocation

Sentence	Speaker ID in the paper	Split
sp01	CH	Development
sp06	DE	Evaluation
sp11	JE	Evaluation
sp16	KI	Development
sp21	SI	Evaluation
sp26	TI	Evaluation

Each sentence contributes two noise types and four nominal SNRs: 48 input cases in total. Development comprises 16 cases; evaluation comprises 32 cases from four other speakers. Only

development signals were enhanced in the local numerical preflight. No enhancement quality scores were computed on the evaluation split before delivery. Inspecting clean/noisy correspondence and validating unity-gain reconstruction did not select parameters by enhancement quality.

The noisy files are the dataset's prepared mixtures of speech and recorded noise. They are not the student's own simultaneous environmental recordings. There is no additional mixing, resampling, time alignment or amplitude fitting in this package. Native int16 amplitudes are read through audioread.

3.1 Initial noise estimate

Both modes estimate the initial noise PSD from the first 0.10 s (800 samples) of each noisy recording. The original demo's 10,000-sample setting is inappropriate for these much shorter sentences. Following the supplied source, the PSD averages 25 ms windowed power spectra at one-sample increments inside this initial interval. These overlapping spectra are strongly correlated; they are not 601 independent noise observations.

The prefix was checked against the corresponding clean signal. It is a **low-clean-energy approximation to a noise-only interval**, not an assertion of exact silence. Across the selected 48 cases, the largest clean-to-residual-noise energy ratio in this prefix is -19.71 dB, about 1.1%. Weak speech or residual recording noise may still contaminate initialization. The same approximation affects both modes. Reference samples are used for this audit only and are never supplied to the enhancer.

4. Implementation and controlled comparison

The supplied WienerNoiseReduction.m implementation is credited to LIU Ming, with modifications by Pascal Scalart, and is available through the MathWorks File Exchange source page. The unchanged source and licence are retained. Its TSNR and HRNR branches are associated with Plapous, Marro and Scalart [2]. The source code is not presented as wholly student-written work.

The original demonstration was run in MATLAB and saved in Scalart_Original_Results.zip. The archive contains the input, TSNR and HRNR outputs, original source, licence and figures. The

saved file identifies MATLAB R2026a Update 5. A separate legacy TSNR check reproduced the saved output to approximately 5.0×10^{-16} maximum absolute difference. This verifies compatibility with that original run; its unpaired demo audio cannot support reference SNR or STOI.

4.1 Shared processing and reconstruction

The comparison uses the TSNR branch of the supplied WienerNoiseReduction.m, associated with Plapous, Marro and Scalart (2006). The main experiment does not compare TSNR with HRNR. Both experimental modes use the same parameters:

Table 2. Shared signal processing parameters

Setting	Value
Sample rate	8 kHz
Window	25 ms / 200 samples; the verified legacy hanning convention
Hop / overlap	80 / 120 samples, or 10 ms / 60%
FFT size	400
Decision-directed smoothing	0.99
Source instantaneous-SNR floor	0.1 after subtracting 1 from power/noise PSD
Source pre-constraint gain floor	0.15
Initial PSD	Same first 800 samples and same estimation function
PSD numerical floor	max(realmin, 1e-12 times mean initial PSD)
Gain constraint	The author's impulse constraint with a 200-sample Hamming taper
Reconstruction	Shared padded overlap-add with window-sum normalization

For reproducibility, the analysis window is $0.5 - 0.5 \cdot \cos(2 \cdot \pi \cdot (1:L)/(L+1))$. The local comparison to the student's actual original MATLAB output verified this convention. Substituting a symmetric zero-endpoint Hann window changes the result.

The author's demo finishes by independently matching each output's peak to the input peak. The quantitative adaptation removes that normalization. Instead, it pads the input by 200 zeros on the left and 400 on the right, accumulates the full inverse-FFT blocks, divides by the accumulated analysis-window weights, and crops back to the original sample range. The same change applies to both modes. Unity gain reconstructs the input to numerical precision. These shared reconstruction changes are recorded adaptations, so the main output is not claimed to be bit-identical to the author's peak-normalized demo. A separate legacy mode verifies the original TSNR branch against the saved MATLAB result.

4.2 VAD guided noise updating

Fixed mode holds the initial PSD constant. VAD mode starts with precisely the same PSD. Frame energy is compared with the current total estimated noise power. A ratio greater than 3 dB gives a speech decision; five further frames are protected by hangover. This is a simple energy detector, not the statistical Sohn detector.

After a frame classified as non-speech, VAD mode updates the PSD for the next frame:

$$D_{\text{next}} = 0.95 * D_{\text{current}} + 0.05 * \text{abs}(Y_{\text{current}}).^2$$

The initial 0.10 s, partially padded frames and frames outside the recording cannot update the PSD. Fixed mode can calculate the same diagnostic VAD decisions but does not use them to alter processing. The 3 dB threshold, five-frame hangover and 0.95 smoothing are fixed engineering choices, not claimed optimal values or parameters copied from Sohn's paper.

The update is applied after processing the current frame, so it changes the noise estimate for the next frame. This timing is recorded in the diagnostic traces. Both modes call the same enhancement function and receive no clean reference. The project contribution is the controlled update option, the shared reconstruction adaptation, the reproducible dataset selection and the evaluation of the resulting behaviour.

5. Evaluation measures and diagnostics

Reference SNR is $10 * \log_{10}(\text{sum}(s.^2) / \text{sum}((z-s).^2))$, where s is the clean reference and z is the unprocessed or enhanced signal. All main SNR/STOI measurements use samples 801 through the end. SNRImprovement_dB subtracts measured input SNR. It measures closeness to the

paired clean waveform, so remaining noise and speech distortion both contribute to the error. It is not a direct measurement of residual noise alone.

STOI is calculated by MATLAB as `stoi(processed,reference,fs)` on the same interval. Larger scores indicate greater predicted intelligibility; scores are not percentages of correctly understood words. An unavailable or failed STOI call produces NaN and a clearly marked partial run. The implementation and range are described in the MathWorks documentation; the research reference is Taal et al. (2011) in the literature review.

The dataset's nominal SNR uses active speech level in its preparation. Our global reference SNR uses the energy in the stated sample interval. Their values need not agree. The generation procedure is described in Hu and Loizou, Section 2.2. The programme keeps both numbers and does not rescale signals to force agreement.

The primary waveform-error measure is defined below, with s representing the paired clean signal and z representing the input or enhanced signal. The sums use samples 801 through the end of the recording.

$$\text{Reference SNR} = 10 \log_{10} [\Sigma s[n]^2 / \Sigma (z[n] - s[n])^2]$$

For each identical noisy input, the paired effect is the VAD score minus the fixed score. Positive values favour VAD for that measure. Evaluation results are reported separately from development. Each noise/SNR condition contains four evaluation sentences, and repeated versions of the same sentence are not independent speakers. No significance test is claimed.

Noise time variation is described by the 90th-minus-10th percentile spread of residual-noise frame power in dB. Residual noise is calculated as noisy minus clean only for analysis after enhancement. Valid 25 ms frames after initialisation are used, with linear percentile interpolation. This descriptor measures amplitude variation across frames and does not capture every form of spectral change.

For activity diagnostics, clean-frame power is measured relative to the maximum clean-frame power in the recording. Frames at or above -30 dB are speech-energy proxies; frames at or below -40 dB are non-speech-energy proxies. The middle band is unlabelled. These proxies are not manually annotated speech boundaries and can miss weak consonants. Noise-update counts

and VAD decisions are therefore diagnostic evidence rather than definitive recognition error rates.

Metrics use unscaled double-precision signals without fitting an extra gain or delay to the clean reference. Listening copies use one shared gain within each clean/input/fixed/VAD quartet if clipping prevention is needed; the applied gain is recorded beside the files. Spectrograms use the same amplitude reference and colour range within each quartet. Filter timings exclude evaluation and plotting and come from one call per method, with fixed always run first. They are not a repeated runtime benchmark.

6. Experimental results

6.1 Overall evaluation results

The following values are arithmetic means over the **evaluation set only**, with 16 cases per noise type and method. These include four sentences at each of the four nominal SNRs. No significance test is claimed.

Table 3. Mean evaluation scores for 16 cases per noise type and method

Noise	Method	Mean reference SNR (dB)	Mean SNR gain over input (dB)	Mean STOI
Car	Unprocessed	7.103	+0.000	0.7959
Car	Fixed TSNR Wiener	10.814	+3.711	0.7964
Car	VAD-updated TSNR Wiener	10.863	+3.760	0.7891
Street	Unprocessed	7.207	+0.000	0.8234
Street	Fixed TSNR Wiener	9.656	+2.449	0.8029
Street	VAD-updated TSNR Wiener	9.800	+2.593	0.8021

Both methods increased mean reference SNR over the unprocessed input. VAD updating produced an additional mean gain of only 0.049 dB for car noise and 0.144 dB for street noise over fixed estimation. Mean STOI decreased by 0.0073 and 0.0008, respectively. The fixed

method's car STOI was close to the unprocessed value, while both street-noise methods had lower mean STOI than the input. These results show that lower waveform error did not consistently translate into higher predicted intelligibility.

Table 4. Paired improvements across the evaluation set

Noise	Cases	Mean VAD minus fixed SNR (dB)	Mean VAD minus fixed STOI	Higher SNR	Higher STOI	Both higher
Car	16	+0.049	-0.0073	7/16	4/16	1/16
Street	16	+0.144	-0.0008	11/16	5/16	5/16

VAD improved SNR in 18 of 32 evaluation cases and STOI in 9 of 32. Only 6 cases improved both measures relative to fixed estimation. All remaining cases are retained in the tables. The comparison therefore does not support a general claim that VAD updating is superior.

6.2 Noise variation and nominal input SNR

Each row below averages four evaluation sentences. Positive values favour VAD updating.

Table 5. Paired differences by noise type and nominal SNR

Noise	Nominal input SNR (dB)	Mean VAD minus fixed SNR (dB)	Mean VAD minus fixed STOI
Car	0	-0.282	-0.0055
Car	5	-0.161	-0.0177
Car	10	+0.731	+0.0004
Car	15	-0.090	-0.0063
Street	0	-0.036	+0.0102
Street	5	+0.213	-0.0063
Street	10	+0.287	-0.0002
Street	15	+0.112	-0.0070

The mean SNR difference was positive for car at nominal 10 dB and street at 5, 10 and 15 dB. The STOI differences followed a different pattern. In particular, car at 10 dB had a 0.731 dB mean SNR advantage but a near-zero mean STOI difference. Street at 0 dB had a positive mean STOI difference despite a slightly negative mean SNR difference.

The evaluated street residual noise had a larger mean frame-power spread than car noise, 8.45 versus 3.02 dB using the P90-P10 descriptor. The scatter plots show mixed positive and negative effects across these spreads. The small set does not establish a monotonic relationship between noise variation and enhancement benefit. Sentence characteristics, initial noise estimation and VAD decisions also vary.

For an individual positive example, sp06/car/10 dB gained 2.027 dB SNR and 0.0160 STOI relative to fixed estimation. Its STOI still remained below the unprocessed input, so this is a relative improvement over the fixed method. In a negative example, sp11/street/0 dB lost 1.054 dB SNR and 0.0106 STOI relative to fixed estimation. These examples were identified after the run to illustrate the range and do not replace the complete comparison.

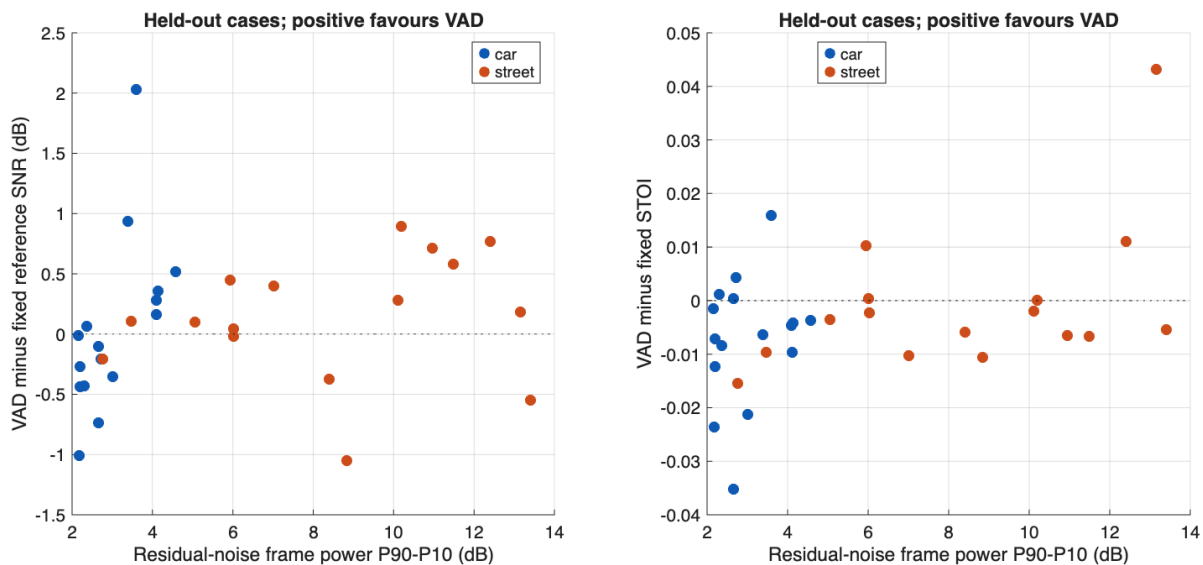


Figure 1. Paired VAD minus fixed effects for all 32 evaluation cases. Positive values favour VAD for the corresponding measure. The horizontal axis describes residual-noise frame-power variation.

7. VAD diagnostics and interpretation

Clean-energy activity proxies provide a limited way to inspect decisions. Frames at or above -30 dB relative to the recording's peak clean-frame power are speech-energy proxies. Frames at or below -40 dB are non-speech-energy proxies. The middle band is unlabelled. These are not manual speech annotations, and they can miss quiet speech sounds.

In the evaluation set, VAD updating used 661 of 3076 speech-proxy frames for car and 664 of 3076 for street, approximately 21.5% and 21.6%. These pooled frame counts show that updates sometimes overlap with reference speech energy. The teacher identified speech entering the noise estimate as a potential failure mechanism. The observed overlap is consistent with that concern, but it does not by itself prove the cause of every STOI decrease.

The preselected sp06/car/5 dB example updated on 23 of 166 speech-proxy frames. The street example updated on 42 of 166. Their diagnostic plots reveal intervals where a non-speech decision overlaps with clean energy. These intervals should be checked during listening. The car VAD plot and street VAD plot are unchanged MATLAB exports.

All five exported figures were visually checked. The spectrogram comparisons share a colour scale within each clean/input/fixed/VAD group. The eight listening WAVs were checked against the saved signals, within their PCM quantization, but no subjective listener rating has been collected. Listening conclusions are therefore still pending.

Other limits are the four evaluation sentences, short recordings, one initial-noise interval and one fixed set of VAD parameters. The nominal SNR files can contain different noise excerpts, so comparisons across SNR levels are not a pure rescaling experiment. The original source's independent output peak normalization was replaced by a shared reconstruction rule in both modes; that adaptation and the complete frozen protocol remain documented.

The observations are consistent with the literature's emphasis on reliable speech-presence decisions [3], [5], [6]. However, the current hard energy gate is simpler than the statistical and minima-based estimators in those papers. It would be inappropriate to infer that MCRA, improved MCRA or statistical VAD would show the same failures. Similarly, the mixed reference-SNR and STOI outcomes are compatible with evaluating waveform fidelity and predicted intelligibility as separate properties [7], [8].

7.1 Spectrogram and listening examples

The following two sp06 examples at nominal 5 dB were selected before the run. Each figure contains the corresponding clean reference, unprocessed input, fixed TSNR output and VAD-updated TSNR output. Colour scales are common within each figure. The images support comparison of time-frequency structure, but a darker output is not sufficient evidence of improved intelligibility. The project website provides the four matched audio versions for each noise type.

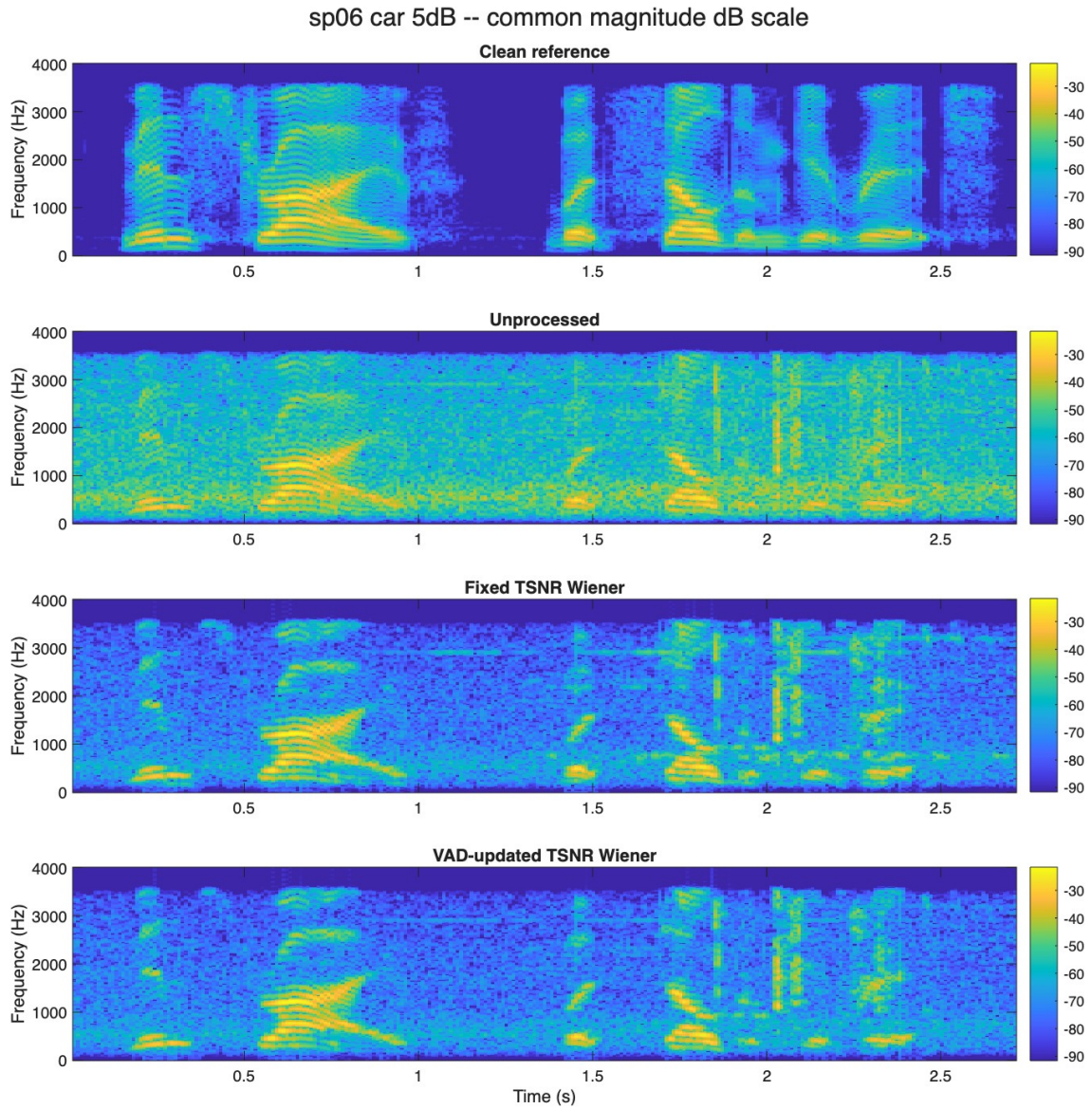


Figure 2. Car noise at nominal 5 dB for sp06, with a common spectrogram scale across the four signals.

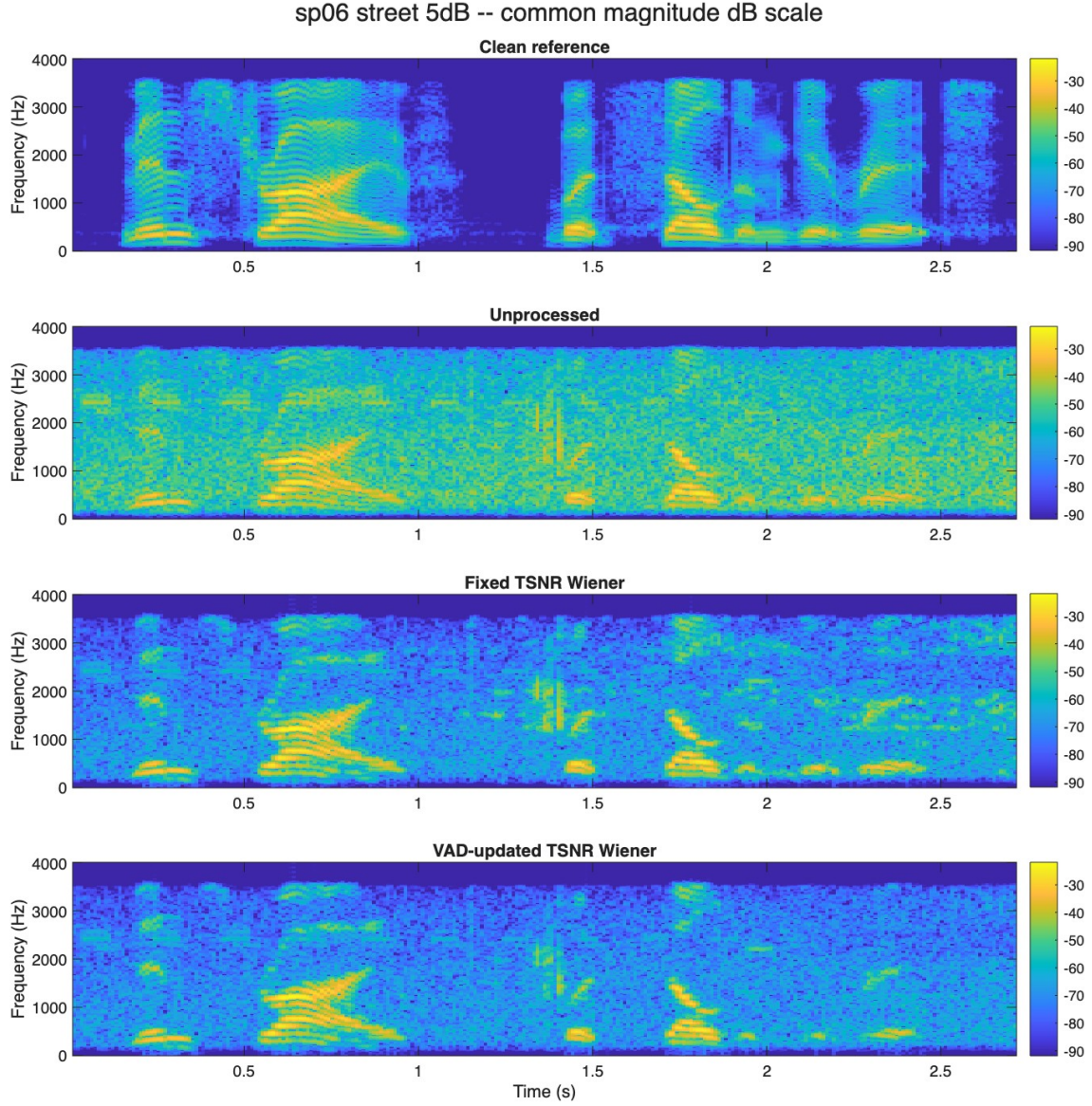


Figure 3. Street noise at nominal 5 dB for sp06, with a common spectrogram scale across the four signals.

7.2 Frame level diagnostic examples

The next plots compare the detector's activity with reference-energy proxies. In the car example, 23 of 166 speech-proxy frames were used for updates; the street example used 42 of 166. These specific intervals should be checked by listening and manual annotation before making a causal claim about speech distortion. Sudden noise may also trigger a speech decision and block an update, so both types of decision error need attention.

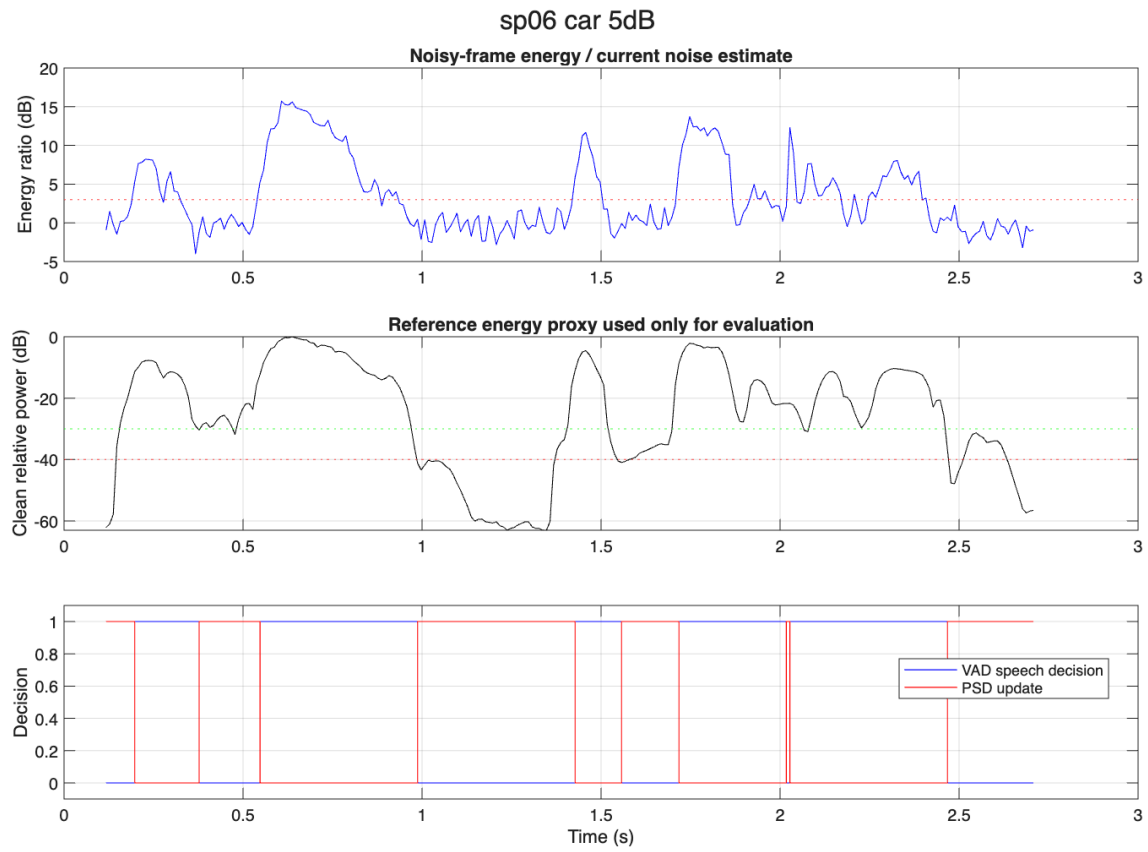


Figure 4. VAD and reference-energy diagnostics for sp06 in car noise at nominal 5 dB.

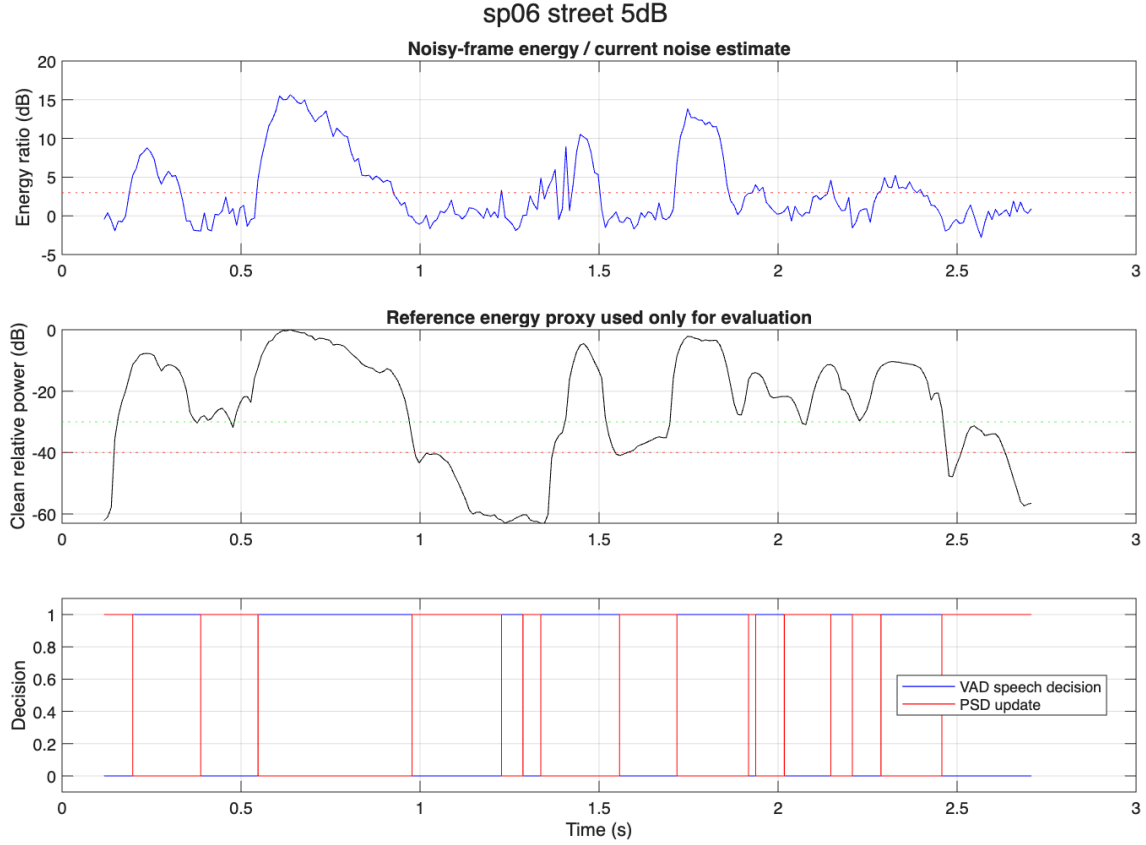


Figure 5. VAD and reference-energy diagnostics for sp06 in street noise at nominal 5 dB.

8. Verification and reproducibility

The run used protocol 2026-09-12-v1 with the result-collection correction 2026-09-13-r1. The correction addressed collection of MATLAB structures into the case table. The native run completed with 48 input cases, 144 metric rows and 144 finite STOI values. The verification gates and result-table regression check passed. Completion establishes that the calculations finished; it does not imply that the VAD method is superior.

Verification covered agreement with the original TSNR output, unity-gain reconstruction, common initial noise estimates, equivalence when noise updates are frozen, and the reference-SNR calculation. The preflight also included silent-input handling and numerical fixtures. Syntax parsing was used as an additional check, and the subsequent native MATLAB run provides the execution evidence.

An independent review read the saved waveforms and reproduced all reported reference SNRs, with a maximum difference below $6e-14$ dB. It also checked corpus sample equality, noise-

update traces, proxy counts, paired differences and condition means. STOI was checked as a completed native MATLAB measurement with finite scores and consistent aggregation; it was not independently recomputed in a second STOI implementation. The review record separates these checks.

To reproduce the run, download Project_Feedback_Two_NOIZEUS with all subfolders, open run_noizeus_experiment.m in MATLAB and run it from the project folder. The package includes 54 selected corpus WAVs and verification data. The recorded run used MATLAB R2026a Update 5; MATLAB Audio Toolbox supplies stoi. The runner creates a timestamped results ZIP and keeps the reviewed run in results/run_20260913_120452. An unavailable STOI call is marked as a partial run rather than treated as a complete result.

The principal outputs and their interpretation are listed below. The original result archive retains all unscaled waveforms, configuration and diagnostic traces. The website also supplies the eight listening WAVs and the unchanged MATLAB figures.

Table 6. Output files and diagnostic definitions

File / column	Meaning
metrics.csv	144 rows: three methods for each of 48 inputs.
condition_summary.csv	Means by split, noise, nominal SNR and method. Evaluation N=4; development N=2. STOI means with missing rows are incomplete.
paired_comparisons.csv	VAD minus fixed SNR and STOI for each identical noisy input. Positive favours VAD for that measure.
dataset_diagnostics.csv	Duration, initialization audit, noise variation and counts behind the proxy diagnostics.
InputWholeRecordSNR_dB	Audit-only whole-recording SNR; its interval differs from the primary measure.
Runtime_s	One filter-call timing, excluding evaluation and plotting. NaN for unprocessed input. No repeated timing benchmark is claimed; fixed always runs before VAD.
PeakAbsolute	Output peak before any listening-copy gain; values above one are kept in the MAT data.

File / column	Meaning
NoiseVariation_dB	P90-P10 of unwindowed residual-noise frame power in dB, on valid 25 ms frames after initialization. Percentiles use linear interpolation at $1+(N-1)p/100$.
InitialCleanToResidual_dB	Clean/noise energy ratio in the first 800 samples; a prefix-contamination diagnostic.
ProxySpeechMissFraction	Fraction of clean-speech-proxy frames given a non-speech VAD decision.
ProxyNonSpeechFalseAlarmFraction	Fraction of clean-non-speech-proxy frames given a speech decision.
UpdateOnSpeechProxyFraction	Fraction of speech-proxy frames actually used in noise updating; zero for fixed mode by design.
NoiseUpdateFraction	Updated frames divided by valid frames after initialization.

Code and corpus attribution are separate. The original source licence remains with the code. The selected WAV files are copied without changing their bytes from the nine NOIZEUS archives supplied for this project, and DATA_AUDIT.json records archive and file hashes. The code licence does not transfer ownership of the audio corpus [8].

9. Remaining work and conclusion

I will listen to the two preselected comparisons and annotate clear speech and non-speech intervals, particularly weak speech that may enter the noise estimate. I will then test a small range of VAD thresholds, hangover lengths and noise-update smoothing values on development data. Any changed configuration will receive a new protocol identifier and appropriate verification fixtures. The existing v1 results will remain visible.

The present evaluation outcomes have already been inspected. If later parameter choices are informed by them, further assessment should use previously unused sentences where possible. Longer noise changes would help test adaptation beyond short corpus utterances. Real fan or traffic recordings remain an optional final demonstration, with their lack of a matched clean reference clearly acknowledged. Listener assessment, a final project report and a demonstration video have not yet been completed.

The completed experiment answers part of the research question. VAD updating increased mean reference SNR slightly relative to a fixed estimate, but reduced mean STOI and improved both measures in only six of 32 evaluation cases. More variable street noise did not produce a uniform advantage. The next stage is to examine when the energy detector's decisions support tracking and when they risk speech suppression, rather than assume that adaptation is always beneficial.

10. Project links

Project website and audio examples

Working code and data

Complete native MATLAB result archive

Detailed results and independent review

Eight paper literature review

The original Proposal PDF remains in the repository as the initial plan. The NOIZEUS controlled experiment and the results described here represent the current quantitative approach following Craig Jin's feedback.

11. References

- [1] P. Scalart and J. Vieira Filho, "Speech enhancement based on a priori signal to noise estimation," in Proc. IEEE ICASSP, vol. 2, pp. 629-632, 1996.
doi:10.1109/ICASSP.1996.543199.
- [2] C. Plapous, C. Marro, and P. Scalart, "Improved signal-to-noise ratio estimation for speech enhancement," IEEE Trans. Audio, Speech, Language Process., vol. 14, no. 6, pp. 2098-2108, 2006. doi:10.1109/TASL.2006.872621.
- [3] J. Sohn, N. S. Kim, and W. Sung, "A statistical model-based voice activity detection," IEEE Signal Process. Lett., vol. 6, no. 1, pp. 1-3, 1999. doi:10.1109/97.736233.
- [4] R. Martin, "Noise power spectral density estimation based on optimal smoothing and minimum statistics," IEEE Trans. Speech Audio Process., vol. 9, no. 5, pp. 504-512, 2001.
doi:10.1109/89.928915.

- [5] I. Cohen and B. Berdugo, “Noise estimation by minima controlled recursive averaging for robust speech enhancement,” *IEEE Signal Process. Lett.*, vol. 9, no. 1, pp. 12-15, 2002. doi:10.1109/97.988717.
- [6] I. Cohen, “Noise spectrum estimation in adverse environments: Improved minima controlled recursive averaging,” *IEEE Trans. Speech Audio Process.*, vol. 11, no. 5, pp. 466-475, 2003. doi:10.1109/TSA.2003.811544.
- [7] C. H. Taal, R. C. Hendriks, R. Heusdens, and J. Jensen, “An algorithm for intelligibility prediction of time-frequency weighted noisy speech,” *IEEE Trans. Audio, Speech, Language Process.*, vol. 19, no. 7, pp. 2125-2136, 2011. doi:10.1109/TASL.2011.2114881.
- [8] Y. Hu and P. C. Loizou, “Subjective comparison and evaluation of speech enhancement algorithms,” *Speech Communication*, vol. 49, pp. 588-601, 2007. doi:10.1016/j.specom.2006.12.006.

Appendix A Complete paired results

Every case is retained below, including development and evaluation. The two score columns give VAD minus fixed differences. Positive values favour VAD. These tables are calculated from the saved `paired_comparisons.csv`; development cases are excluded from the main evaluation means.

Evaluation set

Table 7. All 32 evaluation cases

Case	Noise variation (dB)	SNR difference (dB)	STOI difference
sp06_car_0dB	2.308	-0.431	+0.0011
sp06_car_5dB	4.094	+0.282	-0.0046
sp06_car_10dB	3.587	+2.027	+0.0160
sp06_car_15dB	2.189	-0.271	-0.0124
sp06_street_0dB	10.113	+0.280	-0.0020

Case	Noise variation (dB)	SNR difference (dB)	STOI difference
sp06_street_5dB	3.475	+0.104	-0.0096
sp06_street_10dB	8.397	-0.373	-0.0060
sp06_street_15dB	2.769	-0.208	-0.0155
sp11_car_0dB	4.580	+0.517	-0.0037
sp11_car_5dB	2.651	-0.739	-0.0352
sp11_car_10dB	3.392	+0.938	-0.0063
sp11_car_15dB	4.141	+0.354	-0.0041
sp11_street_0dB	8.844	-1.054	-0.0106
sp11_street_5dB	10.190	+0.897	+0.0001
sp11_street_10dB	12.398	+0.765	+0.0111
sp11_street_15dB	5.048	+0.099	-0.0035
sp21_car_0dB	2.720	-0.207	+0.0044
sp21_car_5dB	4.105	+0.164	-0.0097
sp21_car_10dB	2.651	-0.104	+0.0003
sp21_car_15dB	2.147	-0.009	-0.0015
sp21_street_0dB	5.943	+0.445	+0.0103
sp21_street_5dB	7.020	+0.399	-0.0103
sp21_street_10dB	10.956	+0.713	-0.0065
sp21_street_15dB	6.027	-0.021	-0.0023
sp26_car_0dB	2.181	-1.010	-0.0237
sp26_car_5dB	3.015	-0.352	-0.0212

Case	Noise variation (dB)	SNR difference (dB)	STOI difference
sp26_car_10dB	2.356	+0.064	-0.0085
sp26_car_15dB	2.201	-0.436	-0.0072
sp26_street_0dB	13.152	+0.186	+0.0432
sp26_street_5dB	13.413	-0.550	-0.0054
sp26_street_10dB	6.009	+0.044	+0.0004
sp26_street_15dB	11.483	+0.578	-0.0067

Development set

Table 8. All 16 development cases

Case	Noise variation (dB)	SNR difference (dB)	STOI difference
sp01_car_0dB	2.462	-1.677	-0.0042
sp01_car_5dB	2.791	-0.269	-0.0130
sp01_car_10dB	3.940	+0.315	-0.0051
sp01_car_15dB	2.284	-0.184	-0.0109
sp01_street_0dB	3.371	-0.869	+0.0050
sp01_street_5dB	9.019	-0.019	-0.0003
sp01_street_10dB	12.668	+0.805	+0.0103
sp01_street_15dB	6.363	+0.908	-0.0039
sp16_car_0dB	2.297	-0.461	-0.0233
sp16_car_5dB	2.137	-0.175	-0.0265
sp16_car_10dB	5.403	+0.065	-0.0094

Case	Noise variation (dB)	SNR difference (dB)	STOI difference
sp16_car_15dB	2.781	-0.503	-0.0072
sp16_street_0dB	4.866	+1.363	-0.0033
sp16_street_5dB	5.367	+0.189	-0.0156
sp16_street_10dB	8.088	+1.079	+0.0068
sp16_street_15dB	4.835	+0.316	-0.0034

The complete per-method scores for all 144 rows, development and evaluation condition means, and diagnostic counts remain available in the linked original result archive and CSV files. The appendix lists all 48 paired effects without selecting cases by outcome.